UEI: KULSKCCZJT27	**Agency/ Solicitation: Office of Naval Research, FY2027 DURIP, N0001426SF003**					
	Project Title: High-Repetition-Rate Femtosecond CARS System with a Sub-Atmospheric Reference-Flame Facility					
	Project/Budget Period: 06/18/2027-06/17/2028					
MSU Account Code	**A. Senior Personnel**	**Base Salary**	**Year 1**	**Year 2**	**Year 3**	**Total**
02	PI: Dr. Oluwatobi Busari (0.0 SM)	**$ 105,921**	$ -	$ -	$ -	**$ -**
	B. Other Personnel					
02	Graduate Student (GRA)		$ -	$ -	$ -	**$ -**
02	Technician - none		$ -	$ -	$ -	**$ -**
	Personnel Sub-total		**$ -**	**$ -**	**$ -**	**$ -**
	C. Fringe Benefits					
02	Fringe benefits @ 9%		$ -	$ -	$ -	**$ -**
01 or 02	Fringe benefits @ 42%		$ -	$ -	$ -	**$ -**
	Fringe Benefits Sub-total		**$ -**	**$ -**	**$ -**	**$ -**
	D. Equipment					
11	1. Yb fs amplifier, Light Conversion CARBIDE-40W		$ 220,904	$ -	$ -	**$ 220,904**
11	2. Yb-pumped OPA, Light Conversion ORPHEUS-F		$ 76,164	$ -	$ -	**$ 76,164**
11	3. Dual-channel 4f pulse shaper (LCoS SLM)		$ 35,000	$ -	$ -	**$ 35,000**
11	4. Imaging spectrograph, 320 mm		$ 40,000	$ -	$ -	**$ 40,000**
11	5. High-speed camera, 512 GB (Phantom TMX		$ 264,000	$ -	$ -	**$ 264,000**
11	6. Gated intensifier, Lambert HiCATT25 + gate		$ 152,531	$ -	$ -	**$ 152,531**
11	7. Delay stage, Zaber X-LDQ0300C, 300 mm		$ 8,689	$ -	$ -	**$ 8,689**
11	8. Delay generator, Berkeley Nucleonics 577-8C		$ 7,583	$ -	$ -	**$ 7,583**
11	9. Optical tables, Newport RS2000, with isolators		$ 37,735	$ -	$ -	**$ 37,735**
11	10. Beam optics, power meter, profiler, eyewear		$ 45,000	$ -	$ -	**$ 45,000**
11	11. HEPA/laser-safe enclosure; laser curtains		$ 16,000	$ -	$ -	**$ 16,000**
11	12. GPU acquisition and fitting workstation		$ 12,000	$ -	$ -	**$ 12,000**
11	13. Mod. B: Hencken burner, gas panel (TA-1)		$ 4,690	$ -	$ -	**$ 4,690**
11	14. Mod. B: reduced-pressure vessel (TA-2)		$ 61,300	$ -	$ -	**$ 61,300**
11	15. Mod. B: hydrogen capability (TA-3)		$ 11,500	$ -	$ -	**$ 11,500**
11	16. Mod. B: single-element RDC injector (TA-4)		$ 13,600	$ -	$ -	**$ 13,600**
11	17. Mod. B: spectrometer wavelength ref. (D-1)		$ 14,000	$ -	$ -	**$ 14,000**
11	18. Mod. B: VBG notch filters, OptiGrate (D-2)		$ 16,775	$ -	$ -	**$ 16,775**
	Equipment Sub-total		**$ 1,037,471**	**$ -**	**$ -**	**$ 1,037,471**
	E. Int'l/Domestic Travel					
04	Int'l Travel - PI, acceptance test, Vilnius (1 trip)		$ 3,500	$ -	$ -	**$ 3,500**
	Travel Sub-total		**$ 3,500**	**$ -**	**$ -**	**$ 3,500**
	F. Participant Support Costs					
12	1. Stipends		$ -	$ -	$ -	**$ -**
12	2. Travel		$ -	$ -	$ -	**$ -**
	Participant Support Sub-total		**$ -**	**$ -**	**$ -**	**$ -**
	G. Other Direct Costs					
09	1. Materials and Supplies		$ -	$ -	$ -	**$ -**
09990	2. Publications/Exhibitions		$ -	$ -	$ -	**$ -**
08	3. Consultants		$ -	$ -	$ -	**$ -**
08	4. Computer Services		$ -	$ -	$ -	**$ -**
08191	5. Subaward		$ -	$ -	$ -	**$ -**
12	6. Other - Tuition		$ -	$ -	$ -	**$ -**
08999	7. Other - Event Support		$ -	$ -	$ -	**$ -**
08	8. Other - Extended warranty, laser + OPA		$ 16,320	$ -	$ -	**$ 16,320**
	Other Direct Costs Sub-total		**$ 16,320**	**$ -**	**$ -**	**$ 16,320**
	H. Total Direct Costs		**$ 1,057,291**	**$ -**	**$ -**	**$ 1,057,291**
	Modified Total Direct Costs (MTDC)		**$ 19,820**	**$ -**	**$ -**	**$ 19,820**
15	**I. Indirect Costs (54% MTDC)**		**$ 10,703**	**$ -**	**$ -**	**$ 10,703**
	J. Total Project Costs		**$ 1,067,994**	**$ -**	**$ -**	**$ 1,067,994**

N0001426SF003

Budget Justification

Title: High-Repetition-Rate Femtosecond CARS System with a Sub-Atmospheric Reference-Flame Facility for Time-Resolved Thermometry and Energy-Transfer Kinetics in Naval Propulsion Flows

Budget Period: 06/18/2027 – 06/17/2028

Senior Personnel — TOTAL AMOUNT: $0

PI: Dr. Oluwatobi Busari — 0.0 (SUMR)/yr

Dr. Busari is an Assistant Professor of Mechatronics Engineering at Morgan State University (MSU) and serves as Principal Investigator. He leads specification, procurement, installation, acceptance testing and safety for the instrument. DURIP does not allow direct salaries for faculty, postdoctoral associates or students, so no salary is requested. His base salary is $105,921 for a 9.5-month appointment; 0.0 summer months are requested, for $0 per year. Dr. Busari contributes about 420 hours to installation, integration and acceptance testing at no cost to the award.

A. Senior Personnel	Year 1	Year 2	Year 3	Total
PI: Dr. Oluwatobi Busari	$0			$0
Senior Personnel Subtotal	**$0**			**$0**

Other Personnel — TOTAL AMOUNT: $0

No graduate research assistant stipend is budgeted, because student salaries are unallowable under DURIP. Graduate and undergraduate students contribute about 720 hours to installation, integration and acceptance tests, supported from other sources. No program manager or technician is budgeted.

B. Other Personnel	Year 1	Year 2	Year 3	Total
Graduate Student (GRA)	$0			$0
Technician: none	$0			$0
Other Personnel Subtotal	**$0**			**$0**

Fringe Benefits — TOTAL AMOUNT: $0

MSU fringe charges: For faculty with 9.5-month appointments, fringe charges are budgeted at 42% of the requested salary for the academic year and 9% of the requested salary for the summer. Fringe charges for students and part-time employees are 9% of the requested salary. Because no salary is requested, no fringe benefits are budgeted.

C. Fringe Benefits	Year 1	Year 2	Year 3	Total
PI Busari (SUMR) Fringe @ 9%	$0			$0
Fringe @ 42% (no academic-year salary)	$0			$0
Fringe Benefits Subtotal	**$0**			**$0**

D. Equipment **TOTAL AMOUNT: $1,037,471**

All eighteen items form one instrumentation system for a common research purpose, time-resolved CARS measurement of temperature and molecular state populations, as NOFO Section b.3.i requires. Module A, the 100 kHz fs-CARS diagnostic, is items 1 to 12 ($915,606). Module B, the sub-atmospheric reference-flame facility it measures, is items 13 to 18 ($121,865); it is also the test-article hardware of the PI's pending ONR Long Range BAA proposal, which requests no equipment. All items are purchased at award in Year 1 and are exempt from indirect cost. Items 3, 9, 10 and 11, and all of Module B, are components assembled into the system and are budgeted as equipment; the Office of Research Administration will confirm the fabricated-equipment account. Vendor installation, delivery, and import duty or tariff are part of each item's acquisition cost. Items with a written vendor quotation are budgeted at the quoted price, and the others carry budgetary estimates whose basis is stated below.

1. Yb femtosecond amplifier, Light Conversion CARBIDE-40W-0.4mJ ($220,904). The 1030 nm amplifier (40 W, 400 µJ at 100 kHz, < 250 fs, repetition rate 100 kHz to 1 MHz with an integral pulse picker, chiller and control laptop) is the source of the system. It pumps the OPA and supplies the Stokes beam, and its 100 kHz rate gives one hundred times the measurement rate of 1 kHz CARS instruments. Light Conversion-USA Quotation Q261016-AS1 (19 October 2026, addressed to Dr. Busari, U.S. educational/government discount included): CARBIDE-40W-0.4mJ $190,640; air-freight delivery to the loading dock $3,300; on-site installation and training, including the engineer's airfare and expenses, $7,900. The quotation states that an EU import duty, currently 10 percent, applies and is not included in its total; it is budgeted at 10 percent of the laser price, $19,064. Total $190,640 + $3,300 + $7,900 + $19,064 = $220,904. Delivery is 6 to 8 months after order.

2. Yb-pumped optical parametric amplifier, Light Conversion ORPHEUS-F ($76,164). Converts most of the amplifier output into a tunable 650–900 nm signal (< 55 fs at 800–900 nm; idler 1200–2500 nm) that serves as the broadband CARS pump and probe and reaches the Q-branches of H_2, H_2O, CH_4, N_2, CO and O_2. Same quotation: ORPHEUS-F $69,240, plus the estimated 10 percent import duty, $6,924. Total $76,164.

3. Dual-channel reflective 4f pulse shaper ($35,000). A liquid-crystal-on-silicon spatial light modulator (1272 × 1024) with gratings, optics and fabricated mounts. One channel masks the pump spectrum to select the excited molecules; the other sets a chirped probe of 0.5 to 5 ps. Hamamatsu has identified the X15213-02 (800 ± 50 nm), or its cooled version X15213-02L, as the model for this band and average power, and its quotation has been requested. Budgetary estimate.

4. Imaging spectrograph, 320 mm, aberration-corrected, 1200 l/mm grating ($40,000). Disperses the CARS signal onto the intensified camera (F-mount) with the resolution needed for N_2, H_2 and O_2 Q-branch thermometry. Budgetary estimate based on a public award for a comparable spectrograph (UKRI 2023, £30,453); quotations have been requested from Teledyne Princeton Instruments and Andor.

5. High-speed CMOS camera, 512 GB, Phantom TMX 7510 class ($264,000). Records one CARS spectrum per laser shot at 100 to 200 kHz over a narrow spectral region of interest, and is also used without the laser for high-speed imaging of flames and static fires. Budgetary estimate from U.S. Army purchase order W911QX24P0128 (two units for $527,855); a quotation has been requested from Vision Research (AMETEK).

6. Gated GaAs image intensifier, Lambert HiCATT25, with gate control ($152,531). Adds gain for single-shot spectra down to about 320 Torr and gates out flame luminosity. Tibidabo Scientific Industries (U.S. distributor for Lambert Instruments) Quotation 126946 V2 (20 October 2026, valid to 20 November 2026): HiCATT25/GaAs/3ns/300kHz/1:1/F-mount $135,430; IC-GC-USB gate control $6,376; estimated tariff surcharge $10,725. Total $152,531. Lead time 22 to 26 weeks, Ex-Works Madison, CT.

7. Motorized delay stage, Zaber X-LDQ0300C-AE53D12-KX15C ($8,689). Sets the probe delay (300 mm of travel covers 2 ns). Zaber Quotation SO73573 (9 October 2026): $8,539 plus shipping $150. The sales tax on the quotation ($521.34) is excluded because MSU is tax-exempt. Price held through the June 2027 award.

8. Digital delay generator, Berkeley Nucleonics 577-8C ($7,583). Synchronizes the laser, the intensifier gate and the camera. Quote 33996 (3 September 2026, valid to 3 December 2026): $7,192.50 plus shipping $390.00 = $7,582.50, budgeted as $7,583.

9. Optical tables, Newport RS2000, with pneumatic isolation ($37,735). One table carries the laser, OPA and shaper; the other the CARS beam overlap, the test article and the detection system. Newport (MKS) Quotation Q-296033 (9 October 2026, valid to 15 November 2026), prices including tariff surcharges: RS2000-510-12 $13,446.65; RS2000-48-12 $8,898.17; two S-2000A-423.5 isolator sets $10,860.32; shipping $4,530. Total $37,735.14, budgeted as $37,735.

10. Beam-delivery optics and optomechanics, power meter, beam profiler, laser-safety eyewear ($45,000). Mirrors, telescopes, lenses, dichroic combiners, filters and mounts that route and overlap the three beams, the power meter and profiler used for alignment and acceptance, and Class 4 laser eyewear. Gentec-EO Quotation 69441-1 (14 October 2026) covers the measurement part, $11,338.70 (50 W thermal detector with USB module, BEAMAGE-4M profiler, 60 W attenuator, stands, service plan, ND set, shipping). The remaining $33,661, for Thorlabs optics and Kentek eyewear, is a budgetary estimate with quotations requested.

11. Laser-table enclosure with HEPA fan-filter units and laser-safe panels; laser-safety curtains ($16,000). Protects the amplifier and OPA optics from dust and contains stray beams. Newport Quotation Q-296033 includes the TE80-510 enclosure at $9,436.86. The HEPA units and laser-safe panels have been specified to Newport with a concept drawing, and the free-standing curtains have been requested from Kentek and Rockwell Laser Industries; the balance of $6,563 is a budgetary estimate.

12. GPU acquisition and spectral-fitting workstation ($12,000). Dedicated to the instrument: streams camera data (up to 512 GB per record) and fits spectral libraries to millions of spectra (24-core processor, 256 GB memory, 32 GB professional GPU, 40 TB storage, 10 GbE). Allowable as computing equipment dedicated to DoW-relevant research. Budgetary estimate; a quotation is being obtained through MSU purchasing.

13. Module B: atmospheric Hencken burner, gas panel, pressure instrumentation (TA-1) ($4,690). A 50 mm micro-tube Hencken burner with mass-flow controllers for CH_4, air, O_2 and N_2, pressure instrumentation and a wavelength reference; the atmospheric anchor flame (1600 to 2400 K) for every measurement. Itemized estimate from the PI; the Alicat mass-flow-controller quotation has been requested, and the burner is fabricated in the MSU machine shop.

14. Module B: reduced-pressure vessel and pressure control (TA-2) ($61,300). A 300 mm inner-diameter × 600 mm 304 stainless vessel with seven port flanges ($18,000); six UV-fused-

silica windows ($4,200); a 100 m^3/h chemical-duty dry screw pump ($16,000); a 2 kW exhaust heat exchanger ($2,200); a throttle valve with closed-loop controller ($6,500); capacitance manometers ($4,800); vacuum plumbing ($3,500); burner revisions for reduced pressure ($1,900); and a UV flame detector, interlock PLC and nitrogen purge panel ($4,200). Crystran Quotation APR44409 (16 October 2026) prices the six windows at £7,200 (about $9,635); the item total remains a budgetary estimate until the Kurt J. Lesker (SQ10-0171261) and Busch quotations are received.

15. Module B: hydrogen capability (TA-3) ($11,500). Hydrogen mass-flow controller with CGA 350 two-stage regulator and excess-flow valve ($3,400), two-sensor leak detection with controller ($2,800), a free-standing gas cabinet ($3,500) and a purge panel ($1,800). Applied Energy Systems (Quotation 26957) and PureAire Monitoring Systems (Quotations 1020386 and 1020387) quoted these components on 23 October 2026; the line is held at the estimate until the configuration is selected. The building ventilation for the cabinet is paid by MSU.

16. Module B: single-element rotating-detonation injector (TA-4) ($13,600). A modular injector head in 17-4 stainless steel in three geometries ($8,500), two high-response plenum pressure transducers ($3,200) and a sapphire injector-face window ($1,900). Kulite has given rough-order-of-magnitude prices for its DTL-190 transducer ($1,767 each, five-piece minimum), and its formal quotation has been requested.

17. Module B: spectrometer-based wavelength reference (D-1) ($14,000). A high-resolution fiber spectrometer covering 500 to 1100 nm with Ne, Ar and Hg calibration lamps, the wavelength reference for OPA tuning and for atomic emission lines. It replaces the scanning wavemeter originally considered, which cannot measure broadband femtosecond pulses (Bristol Instruments declined; HighFinesse through TOPTICA confirmed). Budgetary estimate.

18. Module B: volume-Bragg-grating notch filters, OptiGrate BragGrate (D-2) ($16,775). Narrow-band filters at 620, 700 and 780 nm that reject the pump and probe lines and flame chemiluminescence in the collection optics designed by Dr. Albert O. Okorogu. OptiGrate (IPG Photonics) Quotation QD-2026-1014-MGNU-1 (14 October 2026, valid 180 days): three filters at $3,680 each ($11,040), broadband AR coating $5,700, shipping $35. Total $16,775.

D. Equipment	Year 1	Year 2	Year 3	Total
1. Yb femtosecond amplifier, CARBIDE-40W	$220,904			$220,904
2. Yb-pumped OPA, ORPHEUS-F	$76,164			$76,164
3. Dual-channel 4f pulse shaper	$35,000			$35,000
4. Imaging spectrograph, 320 mm	$40,000			$40,000
5. High-speed CMOS camera, 512 GB	$264,000			$264,000
6. Gated intensifier, HiCATT25, with gate control	$152,531			$152,531
7. Delay stage, Zaber X-LDQ0300C	$8,689			$8,689
8. Delay generator, BNC 577-8C	$7,583			$7,583
9. Optical tables, Newport RS2000, with isolation	$37,735			$37,735
10. Beam optics, power meter, profiler, eyewear	$45,000			$45,000
11. HEPA / laser-safe enclosure; curtains	$16,000			$16,000
12. GPU acquisition and fitting workstation	$12,000			$12,000
13. Module B: Hencken burner, gas panel (TA-1)	$4,690			$4,690
14. Module B: reduced-pressure vessel (TA-2)	$61,300			$61,300
15. Module B: hydrogen capability (TA-3)	$11,500			$11,500
16. Module B: single-element RDC injector (TA-4)	$13,600			$13,600
17. Module B: spectrometer wavelength reference (D-1)	$14,000			$14,000

18. Module B: VBG notch filters (D-2)	$16,775			$16,775
Equipment Subtotal	**$1,037,471**			**$1,037,471**

Vendor quotations. The table below lists the quotations behind the equipment and warranty lines (status 4 November 2026). Quotation PDFs are kept with the proposal file.

Item	Vendor and quotation	Date	Valid until	Quoted	Budgeted
1–2	Light Conversion-USA, Q261016-AS1	19 Oct 2026	18 Nov 2026	$271,080 + est. duty $25,988	$297,068
W	Light Conversion-USA, Q261016-AS1, Option 1	19 Oct 2026	18 Nov 2026	$16,320	$16,320
6	Tibidabo Scientific (Lambert), 126946 V2	20 Oct 2026	20 Nov 2026	$152,531 incl. est. tariff	$152,531
7	Zaber Technologies, SO73573	9 Oct 2026	6 Feb 2027; price held to award	$8,689	$8,689
8	Berkeley Nucleonics, 33996	3 Sep 2026	3 Dec 2026	$7,582.50	$7,583
9	Newport (MKS), Q-296033	9 Oct 2026	15 Nov 2026	$37,735.14	$37,735
18	OptiGrate (IPG Photonics), QD-2026-1014-MGNU-1	14 Oct 2026	180 days	$16,775	$16,775
10	Gentec-EO, 69441-1 (power meter, profiler)	14 Oct 2026	30 days	$11,338.70	part of $45,000
11	Newport (MKS), Q-296033 (TE80-510 enclosure)	9 Oct 2026	15 Nov 2026	$9,436.86	part of $16,000
14	Crystran, APR44409 (6 UVFS windows)	16 Oct 2026	30 days	£7,200 (≈ $9,635)	part of $61,300
15	Applied Energy Systems, 26957 (H_2 gas cabinet)	23 Oct 2026		received	part of $11,500
15	PureAire, 1020386 and 1020387 (H_2 leak detection)	23 Oct 2026	120 days	received	part of $11,500
16	Kulite, rough order of magnitude (DTL-190)	14 Oct 2026	n/a	$1,767 each, 5 minimum	part of $13,600
3	Hamamatsu, X15213-02 / -02L	requested	pending		$35,000
4	Teledyne Princeton Instruments; Andor	requested	pending		$40,000
5	Vision Research (AMETEK), Phantom TMX 7510	requested	pending		$264,000
10	Thorlabs optics lot; Kentek eyewear	requested	pending		balance of $45,000
11	Newport HEPA units and panels; Kentek / Rockwell curtains	requested	pending		balance of $16,000
12	Workstation (MSU purchasing)	requested	pending		$12,000
13	Alicat mass-flow controllers; MSU machine shop	requested	pending		$4,690
14	Kurt J. Lesker, SQ10-0171261 (vessel); Busch (pump)	requested	pending		balance of $61,300
17	Spectrometer-based wavelength reference	estimate			$14,000

Travel **TOTAL AMOUNT: $3,500**

One trip in Year 1 for the PI to Light Conversion in Vilnius, Lithuania, to witness the factory

acceptance test of the laser and OPA before shipment, as the NOFO allows for testing and acceptance. Operator training is delivered on site at MSU and is included in the laser quotation. Six days and five nights; airfare is a U.S.-flag or code-share economy fare, in compliance with the Fly America Act.

Purpose	Airfare	Lodging	Taxi / Parking	Meals	Registration	Total
Factory acceptance test, Vilnius	$1,400	$240 x 5 = $1,200	$163	$134 x 4 + $100.50 x 2 = $737	$0	**$3,500**

Lodging and Meals & Incidentals follow the U.S. Department of State per-diem rates for the host city.

E. Int'l/Domestic Travel	Year 1	Year 2	Year 3	Total
International Travel — PI (factory acceptance test, Vilnius)	$3,500			$3,500
Travel Subtotal	**$3,500**			**$3,500**

Participant / Trainee Support Costs **TOTAL AMOUNT: $0**

Participant support costs are not budgeted.

F. Participant Support Costs	Year 1	Year 2	Year 3	Total
1. Stipends	—	—	—	—
2. Travel	—	—	—	—
Participant Support Subtotal	**$0**	**$0**	**$0**	**$0**

Other Direct Costs **TOTAL AMOUNT: $16,320**

Other — Extended warranty, laser and OPA: a one-year extension of the vendor's standard warranty (the second year after shipment), covering parts, labor and on-site service for the amplifier and OPA. Light Conversion Quotation Q261016-AS1, Option 1, $16,320, purchased before the first-year warranty expires. A third-year extension (Option 2, $23,940) is not requested. DURIP allows operation and maintenance costs, including extended warranties. No materials, publication, consultant, computer-service, subaward, tuition or event costs are requested.

G. Other Direct Costs	Year 1	Year 2	Year 3	Total
1. Other — Extended warranty, laser and OPA (2nd year)	$16,320			**$16,320**
Other Direct Costs Subtotal	**$16,320**			**$16,320**

Indirect Costs **TOTAL AMOUNT: $10,703**

Morgan State University's federally negotiated indirect cost rate for on-campus research is 54% of modified total direct costs (MTDC). MTDC excludes student tuition, equipment, and the portion of any sub-award in excess of $25,000. As DURIP requires, all equipment is exempt from indirect cost, so the base is the travel and the extended warranty: 0.54 × $19,820 =

$10,702.80, rounded to $10,703.

Total Project Costs **TOTAL AMOUNT: $1,067,994**

	Year 1	Year 2	Year 3	Total
H. Total Direct Costs	**$1,057,291**			**$1,057,291**
Modified Total Direct Costs (MTDC)	**$19,820**			**$19,820**
I. Indirect Costs (54% MTDC)	**$10,703**			**$10,703**
J. Total Project Costs	**$1,067,994**			**$1,067,994**